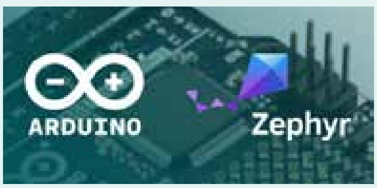
Arduino advances Zephyr RTOS migration

Arduino has rolled out Core v0.3.2, the latest update in its ongoing transition from Arm's now-discontinued Mbed OS to the Zephyr real-time operating system (RTOS). While still in beta, the release delivers critical bug fixes, restores key hardware features, and introduces new capabilities to selected boards.

Zephyr RTOS—an open source, high-performance OS for low-power, resource-constrained devices—offers a scalable and actively maintained platform. Arduino's move ensures continued support for developers while unlocking a richer feature set, including advanced threading, interprocess communication, real-time scheduling, and dynamic sketch loading. The latter compiles Arduino sketches as ELF executables, enabling them to run atop a precompiled Zephyr firmware for improved performance and flexibility.

The migration traces back to Arm's decision to shutter Mbed OS, leaving a gap for boards that relied on it. The Zephyr transition focuses on a subset of hardware previously tied to Mbed OS: Arduino GIGA R1 WiFi, Arduino Opta PLC family, Portenta H7, Portenta C33, and Nano 33 BLE.



By aligning with Zephyr's active, collaborative development model—supported by the Linux Foundation and a broad industry coalition—Arduino positions its hardware ecosystem for sustained innovation, particularly in IoT, automation, and real-time control applications.

integrations, and new developer tools at Ai4 2025. The updates aim to simplify AI workflows, improve accuracy, and lower costs for electronics, embedded, and IoT developers building intelligent applications at scale.

MongoDB's new Model Context Protocol (MCP) server, now in public preview, connects MongoDB deployments directly to popular coding and AI assistants like GitHub Copilot, Claude, and Cursor. Developers can query and manage databases in natural language, streamlining electronics design documentation, manufacturing data analysis, and firmware management.

The company has strengthened its AI partner ecosystem by adding three key players. Galileo brings continuous AI reliability monitoring, ensuring that applications remain accurate and trustworthy in production. Temporal enables the orchestration of durable, scalable AI workflows without the need for complex custom plumbing code, making it easier to build robust systems. LangChain deepens its integration with MongoDB, advancing capabilities like GraphRAG and natural-language queries—tools that are particularly valuable for electronics engineers who require traceable, real-time data in AI-driven workflows.

For AI-enabled electronics—from smart manufacturing systems to predictive maintenance platforms—MongoDB's unified AI stack reduces complexity, enhances performance, and provides a reliable backbone for integrating LLMs, embedded analytics, and real-time sensor data. With its latest moves, MongoDB is positioning itself as a central hub for building high-accuracy, production-ready AI applications—directly relevant to the electronics industry's growing demand for smarter, more autonomous systems.

"Modern AI applications require databases that combine advanced capabilities—like vector search and best-in-class AI models—to unlock meaningful insights from all data types," said Andrew Davidson, SVP of products at MongoDB. "We're giving developers the tools to deploy trustworthy, innovative AI solutions faster than ever."

Perplexity AI eyes the acquisition of Google's Chrome

Indian-origin entrepreneur Aravind Srinivas, co-founder and CEO of Perplexity AI, has submitted an unsolicited all-cash offer worth US$ 34.5 billion (around ₹ 3.02 lakh crore) to acquire Google's Chrome browser—nearly double the AI startup's own valuation.

Aravind Srinivas, co-founder and CEO of Perplexity AI

The proposal, addressed to Alphabet CEO Sundar Pichai, outlines a plan that Perplexity describes as a proactive response to growing regulatory pressure that could eventually force Google to part with Chrome. The offer includes commitments to keep Chromium open source, retain Google as Chrome's default search engine, invest US$ 3 billion (approximately ₹ 26,000 crore) over the next two years, and maintain Chrome's existing workforce to ensure a smooth transition.

According to a term sheet seen by Reuters, Perplexity has proposed to spread the $34.5 billion investment over two years. The company's most recent valuation is $18 billion.

Perplexity is not the only interested party—Yahoo and Apollo Global Management are also reportedly exploring potential bids. The heightened interest